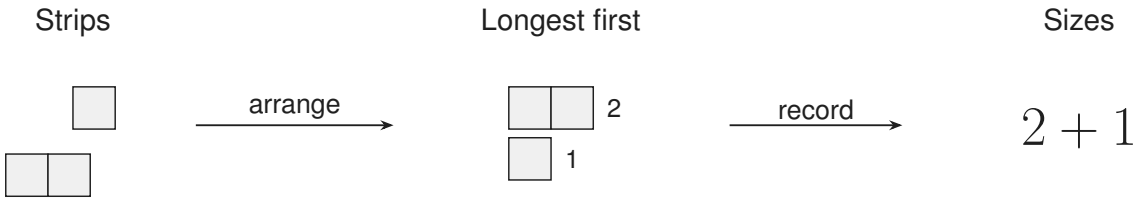
A collection uses all the units, in straight strips. Every strip has at least one unit. Moving or reordering strips does not make a new collection.



Problem 1: Find every collection you can make with 4 units, and with 5 units. Record each once, as rows or sizes.

4 units

5 units

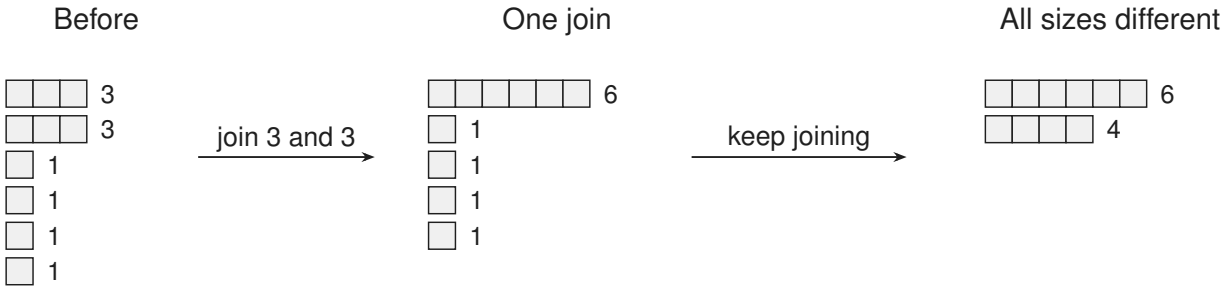
Odd sizes are $1, 3, 5, 7, \ldots$. "All sizes different" means no size repeats.

Problem 2: Choose 6 or 7 units. Find every collection of each kind below.

Total: _____ units

Only odd sizes	All sizes different

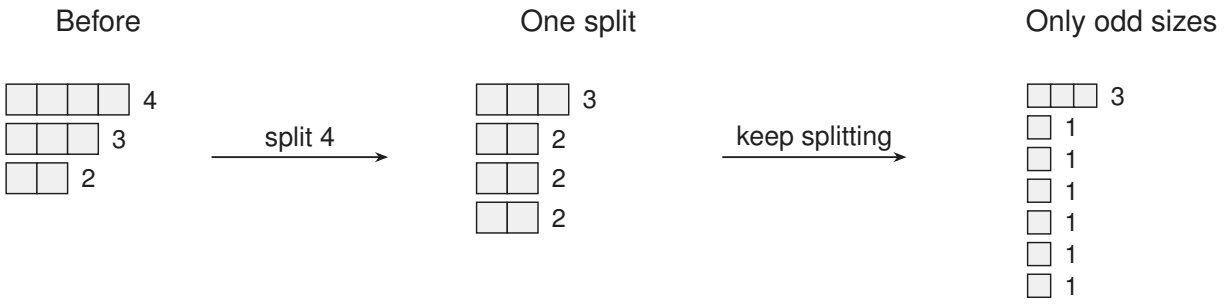
A join replaces two equal-size strips with one strip.



Problem 3: Build each collection. Join equal-size pairs until all sizes are different. Record the final collection.

Before	All sizes different
1, 1, 1, 1, 1, 1, 1, 1, 1	
5, 5, 3, 3, 3, 1, 1	
5, 3, 1	

A split breaks one even-size strip into two equal halves.



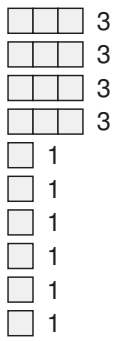
Problem 4: Build each collection. Split even-size strips until only odd sizes remain. Join equal pairs until all sizes are different again. Does each starting collection return?

Before	Only odd sizes	After joining
10 7 4 2 1		
12 6 3		
7 3 1		

Problem 5: With 8 units, find every pair of collections connected by joining and splitting. One collection uses only odd sizes; the other uses all different sizes. Record their sizes.

Only odd sizes		All sizes different
	⟷	
	⟷	
	⟷	
	⟷	
	⟷	
	⟷	
	⟷	
	⟷	

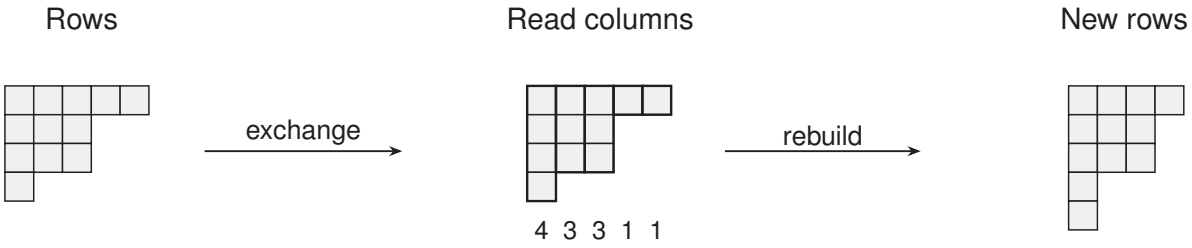
Problem 6: Try different legal joining routes for this collection until all sizes are different. Can the routes end with different collections? Decide whether this can happen from any collection with only odd sizes.



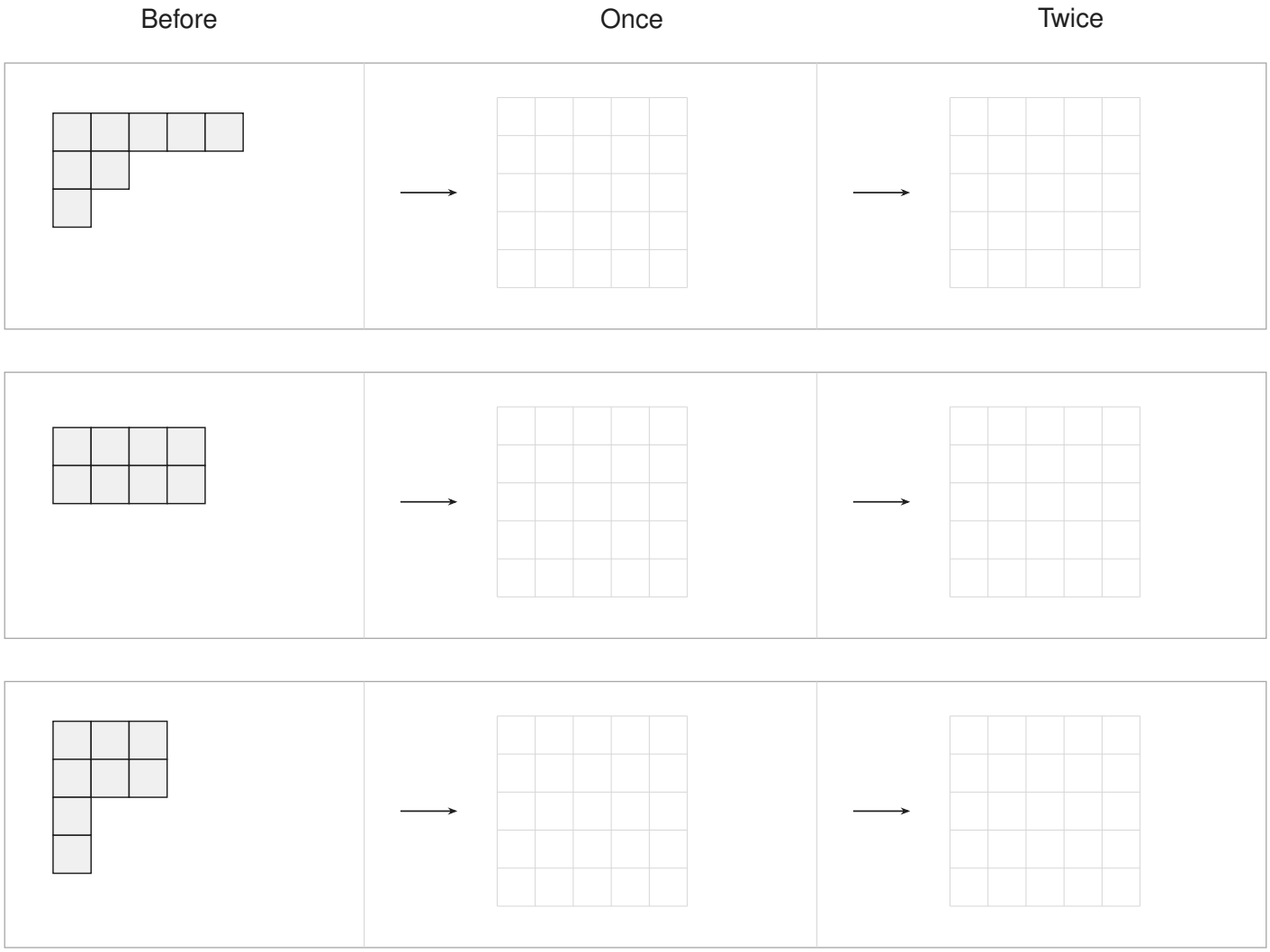
Route 1

Route 2

Put the longest row first, with left edges aligned. To exchange rows and columns, make a new row as long as each column.



Problem 7: Exchange rows and columns twice for each collection. Draw the results. What does the second exchange do?



Problem 8: Find all 8-unit collections with at most 3 strips. Exchange rows and columns, and record their sizes. Which collections can result? Explain a rule that works for any total.

At most 3 strips		After exchanging
	→	
	→	
	→	
	→	
	→	
	→	
	→	
	→	
	→	
	→	
	→	
	→	

Problem 9: For any total, are there exactly as many collections with only odd sizes as collections with all sizes different? Explain how you know.